



Combining process and statistical models for improving forecasts

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- Motivation
- Why combine models?
- A proposal for combination
- Forecasting with combined models

- Considerable flooding across the North of England in December 2015
- 61,000 homes without power for 3+ days
- Widespread damage to infrastructure
- Rainfall and river records broken



- Modelling and forecasting has improved
- BUT it is worst during storm events
- Models often assume things akin to:
independence,
Normal errors,
stationarity



Why combine?

In general:

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- Not try to create a new model, but use what we have and understand already to build confidence and understanding.

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For this project:

1. Create a model for the River Lune.
2. Predict river levels using river gauge data.

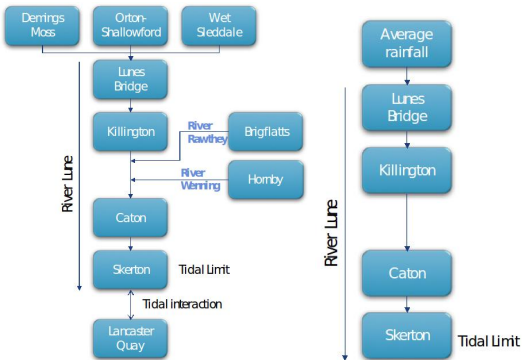
Compare forecasts from:

- Hydrological model widely used in the field
- Hybrid approach with locally stationary residuals

- Use whatever process or statistical model you want
- Take the residuals (model output - data)
- Model the residuals as a non-stationary statistical process
- Use the combination of process and residual forecasting to produce improved forecasts

Nutshell - “Mop up” whatever the model cannot handle to improve forecasts

Case Study: Data

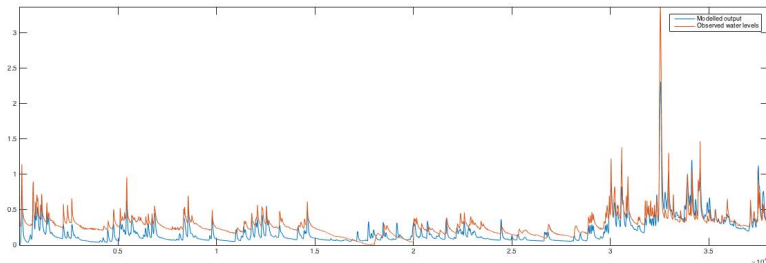


15 minute water level and
average rainfall data.
01/01/2015 to 01/02/2016.

Upstream	Downstream	Time delay
Rain	Lunes Bridge	2 hrs
Lunes Bridge	Killington	4 hrs
Killington	Caton	5 hrs
Caton	Skerton	1 hr
Absolute delay in the system		12 hrs



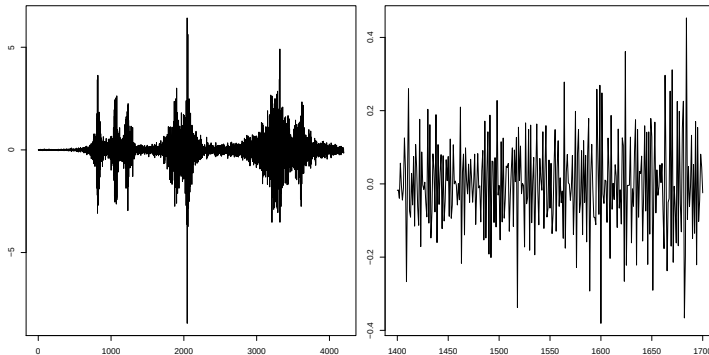
Hydrological model - Cascading transfer function model.



Assumes relationship is constant over time.

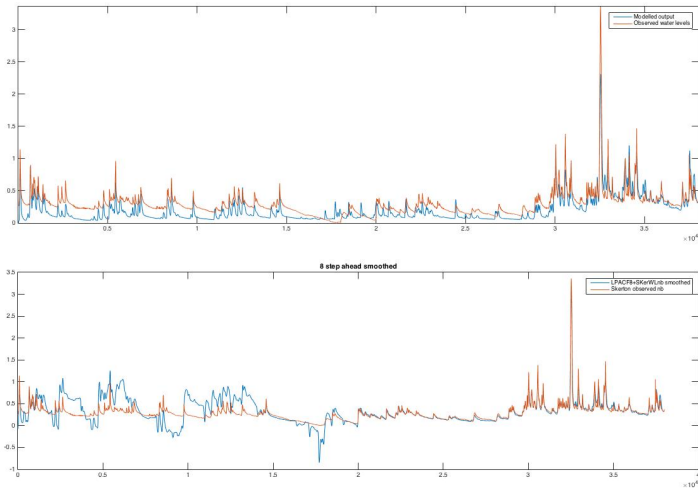


If you take a small enough region, it will appear stationary as the structure varies slowly over time.

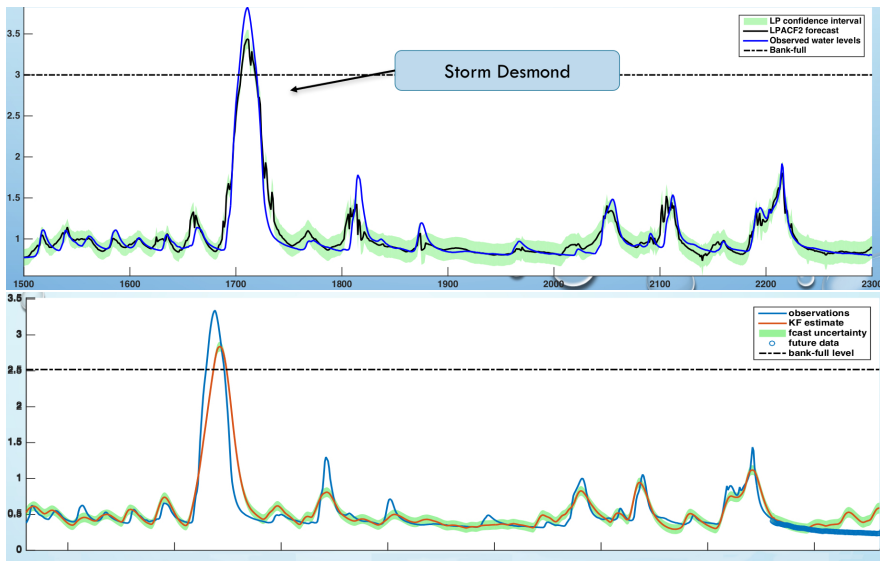


Application areas include;

- medicine, finance
- environmental processes, e.g. wind speeds.



Storm Desmond-2hr lead



- Presented a framework for combining process and statistical models through residuals.
- Created a hydrological model for the River Lune
- Proposed using the Ipfac forecast for river levels
- Showed improvements in forecasting and peak timings
- Further work required on where each method works best
- Need to include the tidal influence in the hydrological model.

Locally stationary wavelet (LSW) processes

LSW processes (Nason *et al.*, 2000):

$$X_{t,T} = \sum_{j=-J(T)}^{-1} \sum_{k \in \mathbb{Z}} w_{j,k;T} \psi_{j,k}(t) \xi_{j,k}, \quad t = 1, \dots, T.$$

- $\{\psi_{j,k}\}$ is a collection of discrete non-decimated wavelets.
- $\{\xi_{j,k}\}_{j,k}$ are zero-mean, orthonormal random variables.
- Wavelet amplitudes $w_{j,k;T}$ controls the non-stationarity.
- LSW processes represent processes whose variance and autocorrelation function vary over time.

The wavelet spectrum is defined as:

$$S_j(z) = \lim_{T \rightarrow \infty} |w_{j, \lfloor zT \rfloor; T}|^2, \text{ where } z \in (0, 1).$$

- Contribution to process variance at scale j and re-scaled time z .
- Estimates are biased and inconsistent – we must correct and smooth.
- As for stationary processes, there is a correspondence between the spectrum and the autocovariance function,

$$c(z, \tau) = \sum_j S_j(z) \Psi_j(\tau).$$